

Probability Misperception in Blind Box Consumption: An Experimental Investigation of Risk and Ambiguity Preferences

Xueting Yang (University of York)

Abstract

Blind boxes, mystery-packaged collectibles sold by firms like Pop Mart, have surged in popularity in China, generating billions in revenue. While marketed as games of chance, their appeal may stem from consumers misperceiving the likelihood of obtaining rare, high-value items. Using a controlled laboratory experiment with the Becker-DeGroot-Marschak (BDM) mechanism, I elicit valuations for blind box-style lotteries to test whether consumers systematically overestimate probabilities. Participants ($n = 102$) evaluated lotteries with a known low-probability "secret" outcome ($q_0 = 0.01$) alongside three equally likely regular outcomes. I compared four theoretical models: *Expected Utility (EU)*, *Rank-Dependent Expected Utility (RDEU)*, *MaxMin (MM)*, and α *MaxMin (AMM)*, finding that EU best explains behaviour (AIC/BIC rankings). Crucially, 65% of subjects overestimated the probability of the secret outcome, with 12% exhibiting statistically significant overestimation (Wald test, $p < 0.05$). Risk-seeking individuals (negative CRRA coefficients) were more prone to this bias. My results suggest that blind box marketing exploits probability misperception rather than ambiguity preferences, offering insights for behavioural industrial organisation and consumer protection policies. These results also have implications for the marketing of insurance products.

JEL codes: D81, D12, C91, D03

Keywords: Blind boxes, probability misperception, risk preferences, ambiguity, BDM mechanism, experimental economics, insurance.

xueting.yang@york.ac.uk

Department of Economics and Related Studies, University of York, Heslington, York, YO10 5DD.

1. Introduction

Blind boxes, mystery-packaged collectibles where consumers purchase without knowing the exact item inside have become a cultural and economic phenomenon in China, particularly among younger generations. Leading toymaker Pop Mart has pioneered this market, leveraging Intellectual Property (IP) like Labubu ("The Monsters") to create artificial scarcity and drive demand. The commercial success is staggering. During China's 2019 "Double 11" shopping festival (akin to Black Friday), PopMart sold 55,000 Labubu blind boxes in 9 seconds (China Daily, 2019). The expansion of blind box market has been rapid. According to Yahoo Finance (2025), the total revenue of Pop Mart increased dramatically by 107% in 2024 compared to the previous year, reaching RMB 13 billion. Figure 1 also displays such overwhelming increasing trend.

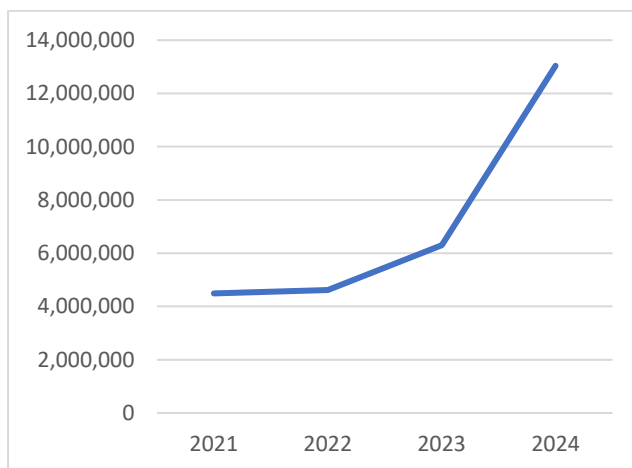


Figure 1. Total revenue of Pop Mart from 2021 to 2024 (Unit: CNY in thousands)

Such rapid ascent of the blind box market represents a significant evolution in consumer economics, moving beyond a mere retail novelty to become a critical case study in value perception, addictive consumption patterns, and the data-driven engineering of desire in the post-digital era. Its analysis is essential for understanding the future of entertainment retail, behavioral market design, and the economic psychology of younger generations. Specifically, it provides an unprecedented empirical window into the behavioural economic forces of uncertainty, anticipation, and addiction in a mainstream commercial setting.

A behavioural puzzle behind the blind box economy is straightforwardly exposed: why do consumers pay premium prices for uncertain outcomes when identical items are available at fixed prices? The answer also has implications for the marketing of insurance products.



Figure 2. A regular figure of Labubu coming from '[The Monsters Big into Energy Series](#)' sold on Pop Mart UK online shop

As the publicity of Pop Mart states, the term “Blind Box refers to packaging that keeps the toy a mystery until opened.” It is not a complete mystery, however, as the toy will be one of an illustrated set. The illustration¹ shows a set of 9 toys (regular figurines) plus one special toy (“a secret figurine”). A consumer can choose to only buy one box from a set or buy a whole set directly. There are 9 blind boxes in one set. If a consumer chooses to buy the whole set, the producer promises that there will be no repetitions, which means that it is possible for the consumer to get all 9 regular figures **or** 8 regular figures **and** the secret figure. The price of a complete set (£121.50) is 9 times the price of a single box (£13.50)².

The consumer is given some ‘information’: “The secret figurine is rarer than the regular figurines”, which suggests that it is more valuable (though there seems no evidence confirming or denying this). For this particular series, they are also told that the “odds of each regular figurine are 1/9” and that “the probability of drawing the secret figurine is 1/108”. I shall use the notation q to represent the probability of the high-value figurine. I note that this ‘information’ is not clear and needs interpretation: if the probability is 1/9 for each of 9 regular figurines, the sum of probabilities cannot be 1, adding in the non-zero probability, q , of the secret figurine. One possible interpretation is that there are two stages: 1st stage: Deciding whether the secret figurine is contained in the box given the probability of having the secret figurine, which is stated as 1/108; If there is no secret figurine, then consumers enter into the 2nd stage: the probability of picking any particular regular figurine in this second stage is 1/9, so the real overall probability of each regular figurine is $(1 - \frac{1}{108}) \times \frac{1}{9}$ which is less than 1/9.

¹There are different series with different numbers of figurines. Here I quote those relevant for the “[Sweet Bean Afternoon Tea Series](#)”.

² This is an oddity in their pricing strategy – probably intended to make it easier for buyers.

Potential buyers of blind boxes were given two possibilities: (1) to purchase a single blind box; (2) to purchase a complete set. In the former case, all that potential buyers were told was that the figurine purchased would be *either* one of the regular figurines *or* the secret figurine. In the latter case, potential buyers were told that the “producer promises that there will be no repetitions”, which means that it is possible for the consumer to get all 9 regular figures **or** 8 regular figures **and** the secret figure. The price of a complete set (£121.50) is 9 times the price of a single box (£13.50). I do not investigate subjects who choose to purchase a complete set as that is an entirely different decision problem³.

As above mentioned, it is not clear how this ‘information’ is intended to be, or might be, interpreted. This opacity suggests two potential drivers of demand: (1) Rational ambiguity-seeking (e.g., preference for unknown probabilities); (2) Systematic probability misperception (e.g., overestimation of rare outcomes. Prior work attributes blind box popularity to probability distortion (Mvondo et al., 2023; Prospect Theory), pleasure of unboxing (Bonniot-Cabanac & Cabanac, 2009) and Impulsive consumption (Zhang et al., 2022). However, no study has experimentally isolated probability misperception while controlling for these confounds. My research addresses this gap by testing:

- 1) Do consumers systematically misperceive the probability of obtaining high-value items in blind boxes?
- 2) Does this misperception explain their willingness-to-pay, even under full transparency?
- 3) Are standard models (e.g., Expected Utility) sufficient to explain behaviour, or do ambiguity models add predictive power?

According to the lab experiment, this research found that consumers persistently overvalue lotteries and misperceived their chances even when objective probabilities are disclosed. This demonstrates that the bias is not rooted in ignorance, but in the affective and cognitive processing of the low-probability event itself. This finding has direct and profound implications for understanding failures in protective markets like insurance. It demonstrates that the mere provision of statistical information like a common regulatory remedy is often inadequate to correct biased decision-making under uncertainty. Just as a blind box purchaser overlooks disclosed 1% odds in the thrill of the potential 'hit,' an individual may discount the disclosed 0.1% annual probability of a disaster as 'negligible,' failing to internalize its catastrophic potential. In both cases, the objective statistic is overridden by subjective, affective valuation. This explains why simply mandating transparency in insurance documents may do little to close the 'protection gap' for low-probability, high-impact risks.

2. Methodology

To answer these questions, I conducted a laboratory experiment which explicitly asked subjects to value a number of different lotteries (some of which were essentially blind boxes⁴). I used the Becker-DeGroot-Marschak (BDM) method (Becker *et al*, 1964), to elicit valuations. I implemented the experiment in two parts. In Part 1 participants were asked to value lotteries with three possible known

³ Also, an experiment would be very costly!

⁴ For perhaps obvious reasons I simplified the blind boxes: instead of having 9 or more regular figurines, I just had 3; I made them all equally valuable and made the value of the secret figurine much higher than the regular figurines. My experiment captures the essence of blind boxes, without confusing the subjects with too much detail.

and different risky outcomes ($O_1 < O_2 < O_3$). From their responses I could infer each subject's risk aversion and their weighting function parameter⁵. In Part 2 subjects were asked to value a set of four-outcome (possibly ambiguous) lotteries. The lotteries in Part 2 differed from those in Part 1 through the addition of a large low-probability outcome - £200 (denoted by O_4) with a very low chance of 0.01 (denoted by q_0). This represents the secret/hidden figurine. I then combined their responses in Part 2 with their previously elicited risk preferences from Part 1, to infer the probabilities that subjects appeared to be attaching to the various outcomes. Particularly, I could infer their subjective probability for the high-value outcome. This helps to understand whether their perception of the probability of getting the high-value outcome is greater than or smaller than the true probability.

To explain subjects' behaviour, my strategy was the following: I started by fitting to the data (the valuations in Part 1 and Part 2) four well-known economic theories of decision-making under ambiguity (and risk), namely Expected Utility (EU) theory, Rank-Dependent Expected Utility theory (RDEU), the MaxMin Model (Gilboa & Schmeidler, 1989) and the α MaxMin Model (Ghirardato *et al*, 2004). I included both EU and RDEU versions of MaxMin and α MaxMin. I then ranked the explanatory power of these various models (using both the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC)), hence determining which were the better stories to explain the data.

In order to fit the various models, I needed to make assumptions about the form of the utility function (for all the models) and the form of the probability weighting function (for RDEU and its variants). For the utility function I assumed that it was of the CRRA form:

$$u(x) = \frac{x^{1-r}}{1-r}, r \neq 1; u(x) = \log(x), r = 1,$$

Here the parameter r indicates the subject's risk aversion.

For the weighting function in RDEU I took the commonly-accepted form $w(p) = \frac{p^g}{[p^g + (1-p)^g]^{1/g}}$ proposed by Kahneman and Tversky (1979). The parameter g reflects the degree of probability distortion.

The procedure I adopted with all the models was the same. I used the subject's valuations in Part 1 to estimate r for all the models and g for RDEU and its variants. I then used the valuations in Part 2 (combined with each subject's estimated r , and g where appropriate) to estimate each subject's perceived probability of the high outcome). I used Maximized Likelihood Estimation throughout. As I have already noted, the target of the estimation was to estimate the value of q (the probability of the high-outcome event) apparently used by each subject. According to the result of Akaike's Information Criterion (AIC) comparison, the EU model seems to fit the data best with the RDEU model fitting slightly worse than the EU model; then came the MaxMin model, with the α MaxMin Model was the worst. The result of the comparison of Bayesian Information Criterion (BIC) was slightly different with that by using the AIC, and the MM-EU model replaced the RDEU model to be the second-best, but more importantly the EU model still emerges to fit data best. For over half of the subjects, with the

⁵ How they would be parameterised will be explain later.

EU model, the perceived probability of O_4 happening, q , was greater than its objective probability q_0 , so consumers may pursue blind boxes because they misperceive the probability of the largest possible outcome happening and thus overvalued this outcome.

3. Related Literature

According to recent existing studies, reasons for blind box consumption behaviour are mainly divided into three: (1) probability misperception or distortion; (2) pleasure from unboxing process and collection; and (3) impulsive purchase.

Possible theories to explain probability misperception or distortion include Prospect Theory (Kahneman & Tversky, 1979), the Availability Heuristic (Tversky & Kahneman, 1973), and the Gambler's Fallacy. Prospect Theory stresses that individuals mis-weight probabilities of possible outcomes, overweighting small probabilities and underweighting large probabilities. Considering the blind box context, individuals may overestimate the probability of getting the secret figure. Prospect Theory also characterises what is termed the *reference point*; Chen (2024) argues that in the blind box context, obtaining the rare secret figure may be considered as the reference point, and hence any other outcome is considered a loss relative to this. This makes them more risk-loving, and explains why they keep buying blind boxes.

Mvondo et al. (2023) applied Prospect Theory to explore the blind box selling strategy based on survey data. According to Kahneman and Tversky (1979) Prospect Theory, individuals would be risk-seeking for losses and risk-averse for gains. Also, due to the loss aversion indicating that individuals experience significantly greater pain from losing a sum of money than pleasure from gaining an equivalent amount, when consumers failed to get the secret figure, they treated this result as a loss, and the chance of getting that figure stimulated them to buy more boxes, reflecting a risk-seeking behaviour.

The Availability Heuristic postulates that the perceived probabilities for individuals depend on how easily similar cases come to the individual's mind. There appear to have been few empirical studies that have explored whether this factor really affects consumers' probability perception in the blind box market. However, with experiments, Hertwig et al. (2004) compared decisions from description and those from experience and found that people underweight rare events when learning from descriptions like the stated probability but overweight them after exposure to examples.

The Gambler's Fallacy, first proposed by Tversky and Kahneman (1971), stresses the cognitive bias of individuals who wrongly believed that past random events could affect future probabilities in independent trials. An empirical study conducted by Xia et al. (2025) demonstrated that the Gambler's Fallacy drove the misjudgement of probability. They mistakenly formed cognitive expectations and overestimated the probability of getting the secret figure next time.

The above studies can explain why individuals keep buying blind boxes, but few of them answered the question: what if individuals misperceived probabilities from the beginning? Experimental investigations in this area are rare. My study implements a one-shot experiment, excluding factors such as the Availability Heuristic and the Gambler's Fallacy, and aims to answer these questions: are consumers necessarily risk-seeking? Are individuals allured by the large prize itself and thus overestimate its probability of happening?

It should be noted that my experiment uses lotteries rather than real blind boxes. Consequently, it avoids some features of blind box consumption such as the pleasure from the unboxing process and collection (e.g. Bonniot-Cabanac & Cabanac, 2009; Mvondo et al., 2023; Phan & Hoai, 2025) and impulsive purchases (e.g. Zhang et al., 2022; Gong et al., 2024; Xia et al., 2025).

I can conclude that there have been very few experimental studies investigating blind box consumption behaviour, and certainly none from the perspective of individual decision-making under risk and ambiguity. This experiment was designed to shed light on understanding individuals' risk and ambiguity preferences and behaviour *in the blind box context*. By estimating the perceived probability of the largest possible outcome (which is the 'secret figurine' in the blind box), my experiment provides an explanation in the setting of blind boxes: sellers often manipulate scarcity perceptions (e.g., "limited edition" or "hidden/secret figure"). It answers the question: does this setting inflate consumers' valuations for blind boxes through an inflated probability of getting the hidden/secret figurine?

4. Experimental Design

This laboratory experiment involved four sessions with 102 subjects in total. It was conducted at the University of York; the participants were undergraduate and postgraduate students and members of staff. Sessions lasted on average 45 minutes; the average payment was £24.26 including a £2.50 show-up fee. The crucial method used in the experiment was elicitation of valuations. The BDM method, a common way to elicit the value of an item to an individual, was used throughout. Its importance in the experiment merits a discussion of it.

4.1 The BDM method

I am interested in the subjects' valuations of lotteries. I decided to use the *seller's version* of the BDM mechanism (eliciting their Willingness to Accept, or WTA) as the way of measuring this. I could, of course, have used the *buyer's version* (eliciting their Willingness to Pay or WTP). I decided against this latter because of the practical problems of implementing the buyer's version. These would have arisen because the lotteries in Part 2 of the experiment contained a large low-probability outcome. First, the experiment would have been much more expensive if subjects were given an endowment equal to that large outcome. If, instead, subjects were provided with an endowment less than that large outcome, it is possible that they would face a real loss if they were allowed to state a bid price higher than the endowment, and because of loss aversion, their bid price would be limited by the endowment and would not state their true valuation. The sellers' version helped to avoid such problems.

As noted above, my experiment uses the Becker-DeGroot-Marschak (BDM) method to elicit valuations. I need these valuations in order to estimate key parameters (r , g , α and q). For some of these parameters there are other ways to estimate them. There are some studies which compare the different methods (e.g. Dave et al., 2010; Asioli et al., 2021). These other methods include multiple lists (Holt & Laury, 2002), Pairwise choices (Hey & Orme, 1994) and the allocation method (Loomes, 1991). No method emerges as clearly the best, so I chose one which was simple for subjects to understand.

Let me describe the method as implemented in Part 1. The method involves a series of decision problems, in each of which subjects are presented with a lottery with three possible outcomes ($O_1 < O_2 < O_3$) and their corresponding probabilities p_1 , p_2 , and p_3 . Participants are asked to imagine that they own this lottery. They are asked to state their minimum WTA, that is the lowest price at which they are willing to sell this lottery. They were asked to state a value between and including 1 and O_3 . Denote their stated value by V . The method is 'played out' as follows. After a subject has stated a valuation for all problems, the computer would randomly choose one of the problems and retrieve its corresponding lottery and the participant's stated value. A random number, $x \in [1, O_3]$, would then be selected by the computer to be the bid price for the lottery. The computer here can be seen as a buyer. There would be two possible implications:

- a. If $V \geq x$, the participant will sell the lottery and get x as the payment.
- b. If $V < x$, there is no transaction, and the participant would need to play out⁶ the lottery. The outcome of the playing out would be the payment for this participant.

Figure 3 is an example displaying one of the problems in Part 1 of the experiment. For this case, the participant owns a lottery which implies an outcome £1 with a 40% chance, an outcome £6 with a 30% chance, an outcome £12 with a 30% chance. In the experiment, a participant could state his/her valuation by moving the slider in which the minimum value is 1 and the maximum value is 12. The horizontal line in the graph indicates the stated value and changes as the subject moves the slider. In this example, the stated value is 4.2 which is the minimum price this participant is willing to sell.

If this problem is selected by the computer, the computer will randomly generate a bid price, x , between 1 and 12. If x is 6, that is greater than 4.2, this bid price also would become the transaction price, and the participant would receive £6 as his/her payment for this part. If x is 3, that is lower than 4.2, the participant would play out this lottery, so the payment for this part would be, so the payment for this part would be £1 with a chance 40%, £6 with a chance 30% and £12 with a chance 30%.

⁶ 'Playing out' involves a random implementation of the lottery.

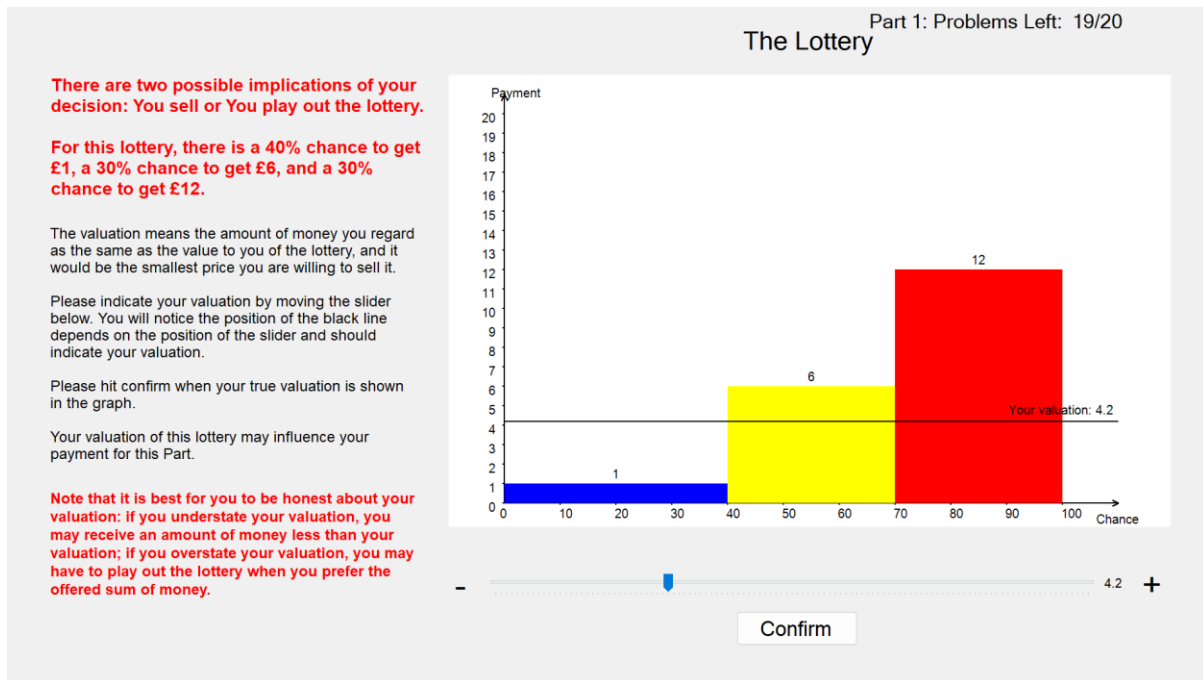


Figure 3. The software interface in Part 1

4.2 Experimental procedure

The experiment was divided into two parts: in Part 1, participants were asked to value 20 three-outcome lotteries; in Part 2, they were asked to value 20 four-outcome lotteries. The main difference between the lotteries in Part 2 and those in Part 1 was the addition of a large low-probability outcome (the 'secret figurine').

It should be noted, that before the experiment, the lotteries and their parameters, like the possible outcomes and their corresponding probabilities for each lottery, were chosen after simulation with MATLAB. The simulation enabled me to check whether the lotteries were able to estimate with sufficient accuracy the key parameters (r , g and q) in which I was interested. The simulation also helped to check whether the theoretical models were identifiable using Maximum Likelihood method.

All the lotteries in Part 1 of this experiment had three possible outcomes: O_1 , O_2 and O_3 ($O_1 < O_2 < O_3$) with corresponding probabilities: p_1 , p_2 and p_3 . These outcomes and probabilities changed over problems. The only thing participants were asked to do was to express their valuation of each lottery, (described to them as the smallest amount of money for which they would be willing to sell the lottery). The software interface for this part is displayed in Figure 3. To help participants better understand the lottery, I displayed the lottery for each problem using the graph on the right side of the interface. The width of a rectangular indicates the probability of a possible outcome happening, and the height represents the amount of money for that possible outcome. Three colours: Blue, Yellow and Red were used to represent the three possibilities for each lottery. Participants could show their valuation by moving the slider so that the height of the horizontal black line in the graph was equal to their valuation.

The setting of Part 2 is similar to that of Part 1, but the structure of lotteries in this part is different. All the lotteries have *four* possible outcomes, one (the ‘secret figurine’) considerably larger than the others. In this part, the probabilities of all possible outcomes and the amount of the largest possible outcome were fixed. Only the amount of the payment for three possible outcomes changed over problems. To capture the feature that the probability of getting all regular figurines is equal and that of getting the secret figure is very small, in our experiment, the largest possible outcome, O_4 , always equalled £200, with chance $q_0 = 1\%$, and the probability of any one of the other three possible outcomes happening equalled to 33%, that is, $p_1 = p_2 = p_3 = 33\%$. Figure 4 shows the software interface in Part 2. As in Part 1, participants were asked to move the slider to indicate their valuation for each lottery.

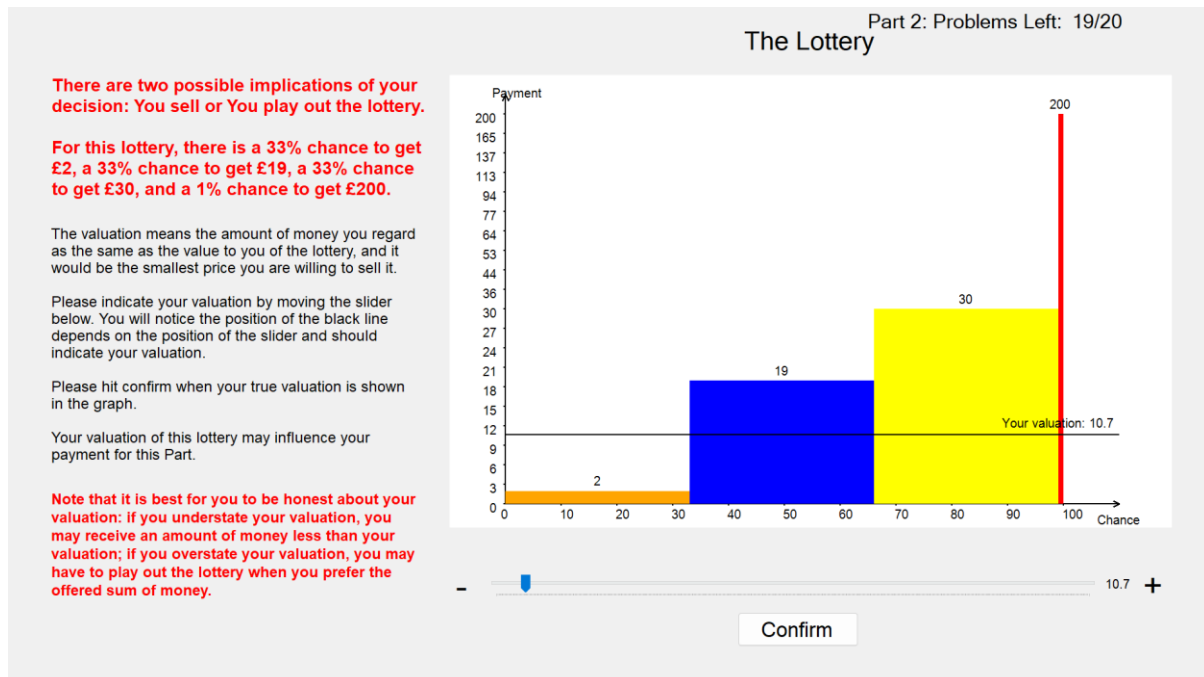


Figure 4. The software interface in Part 2

In each Part, after valuing all 20 lotteries, one was selected by the computer and the following procedure would be followed. The computer chose a random number between and including 1 and the highest outcome in that lottery (200). The random numbers chosen by the computer were shown in the experimental software. An example is displayed in Figure 5. If that number is greater than a participant’s valuation, their payment for this part will be the random number; if that number is less than the valuation, the participant will play out the lottery and the payment for this Part will be the outcome. This BDM mechanism is a powerful way to encourage participants to state their true valuation: if they understate their valuations, they may get an amount of money less than their valuations; if they overstate their valuations, they may have to play out the lottery when they prefer the offered sum of money.

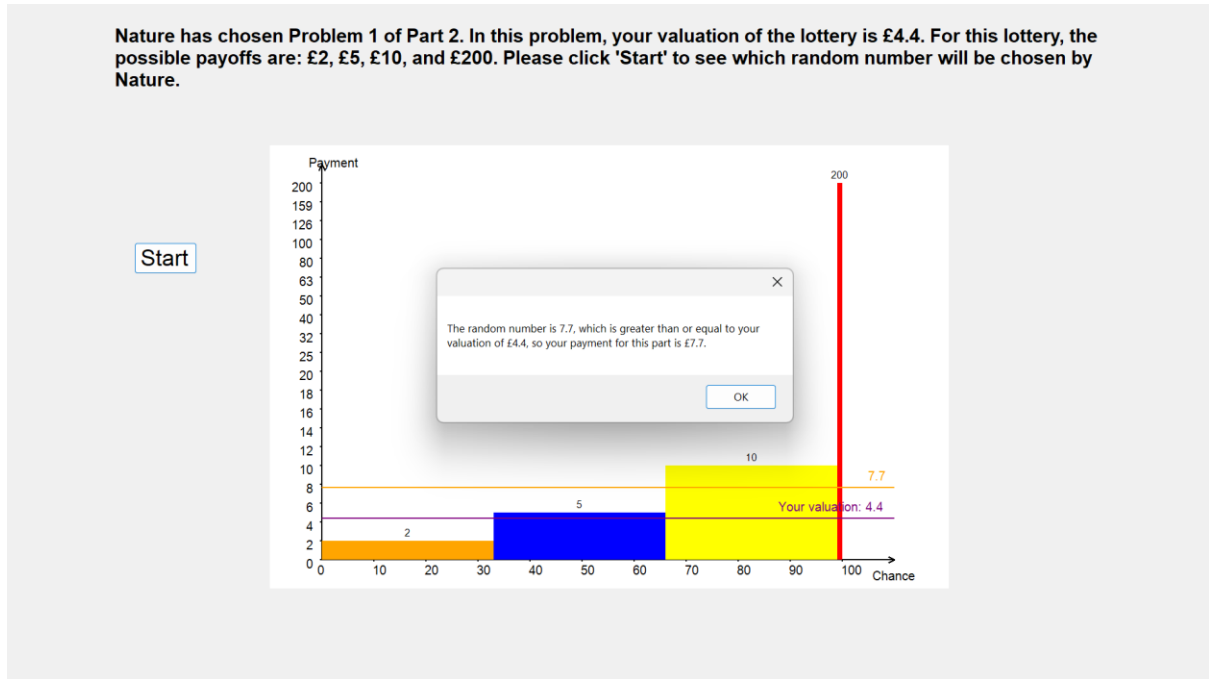


Figure 5. The software interface of playing out a random number for a problem in Part 2.

5. Theoretical Models

I considered as possible explanations lying behind participants' decisions the following models: Expected Utility theory, Rank Dependent Expected theory, the α MaxMin Model, and the MaxMin model (and variants of the latter two under both EU and RDEU). The first two are models of behaviour under risk; the last two are models of behaviour under ambiguity. I note that these four models are just a subset of numerous theories of individual decision-making under risk or ambiguity. Others like the Smooth Model (Klibanoff et al., 2005), the Choquet Expected Utility model (Schmeidler, 1989) and the Variational Preferences (Maccheroni et al., 2006), are more complicated with many parameters that need to be estimated. Behaviourally, they seem difficult for human beings to implement. The key point for this research is to estimate the perceived probability⁷ q of O_4 , the largest possible outcome for lotteries in Part 2, and all other parameters serve for the estimation of q . To concentrate on this key point, these more complicated models were not investigated. Perhaps they should in future research, but my preliminary results suggest that simplicity is crucial.

For each model, the Certainty Equivalent (CE) of each lottery is required to be calculated first, expressed in terms of its parameters. This is used by Maximum Likelihood to estimate the parameters from the stated valuations. The process of the calculation of CE is shown in the technical appendix.

6. Econometric Methods

Four models (and two sub-models) are fitted to the experimental data by using Maximum Likelihood: Expected Utility (EU); Rank Dependent Expected Utility (RDEU); the *MaxMin* (MM) model (with two

⁷ I denote the *true* probability of the hidden figurine by q_0 , to distinguish it from the *perceived probability* q .

sub-models: the MM-EU model and the MM-RDEU model); and the α MaxMin (AMM) model (with two sub-models: the AMM-EU model and the AMM-RDEU model). The data used in the estimation are the participants' stated values for lotteries: 20 for Part 1 and 20 for Part 2. To add in a stochastic component, I assumed that these valuations, V_i ,⁸ were stated with an additive error ε that was normally distributed with mean 0 and standard deviation σ , so $V_i = CE_i + \varepsilon$ where ε is $N(0, \sigma^2)$

The main target of this part is to estimate participants' perceived probability of the largest possible outcome O_4 happening in Part 2. By applying these four models, I can check whether participants misperceived probabilities when they faced an attractive big prize within a lottery although they were informed of the real probability of that outcome happening. The objective probability of O_4 was denoted by q_0 , and the perceived probability was denoted by q . In order to estimate q , I needed to estimate r and g first. According to valuations from Part 1, I estimated parameters r and g for the RDEU model with σ_r , parameters r and σ_r for both EU, MM and AMM models. Based on these estimated parameters and observations in Part 2, the values of perceived probability q could be inferred.⁹ For the MM model, participants were assumed to only consider the lowest perceived probability $\underline{q}$ when they valued lotteries. The AMM participants would identify both the lowest and largest perceived probabilities: $\underline{q}$ and $\bar{q}$, and take a weighted average on the worst and best expected utility with the parameter α that is the weight on the worst expected utility. Table 1 summarizes the parameters estimated, the number of them for each model, and observations from that part that were used in estimation.

Model	RDEU	EU	MM		AMM	
			MM-EU	MM-RDEU	AMM-EU	AMM-RDEU
Number of parameters	5	4	4		6	
Part 1	(r, g, σ_r)	(r, σ_r)	(r, σ_r)		(r, σ_r)	
Part 2	(q, σ_q)	(q, σ_q)	$(\underline{q}, \sigma_q)$		$(\underline{q}, \bar{q}, \alpha, \sigma_q)$	

Table 1. The summary of parameter estimation

7. Results

The detailed estimation results for each model and corresponding log-likelihoods can be found on the [EXEC website](#). To compare the four models mentioned above, both the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) were computed. The former is calculated by $AIC = 2k - 2LL$ and the latter by $BIC = \ln(n)k - 2LL$, where LL is the maximised Log-Likelihood, and where k indicates the number of parameters of a model, and n is the sample size ($n = 102$ subjects in my experiment). A single maximised log-likelihood used in the calculation was the sum of the maximised log-likelihoods for Part 1 and Part 2. The model rank results using these two criteria are shown below:

AIC Rank:

⁸ i is the lottery number.

⁹ Note that the sequential procedure I have used means that the estimate q of q_0 takes into account each subject's individual risk preferences. If, instead, I assumed risk neutrality and linear weighting of probabilities, I would have obtained distorted estimates of q . Moreover, it is obvious that one cannot estimate the risk preferences and the perceived probability simultaneously because of identifiability issues.

	Best	Second best	Third best	Fourth best	Fifth best	Worst
EU	31	17	14	31	8	1
RDEU	25	29	22	21	4	1
AMM-EU	1	2	3	2	34	60
MM-EU	22	17	6	20	36	1
AMM-RDEU	2	3	29	12	17	39
MM-RDEU	21	34	28	16	3	0

BIC Rank:

	Best	Second best	Third best	Fourth best	Fifth best	Worst
EU	42	22	25	9	3	1
RDEU	18	25	23	31	4	1
AMM-EU	0	3	2	1	50	46
MM-EU	26	29	4	23	19	1
AMM-RDEU	0	3	8	14	24	53
MM-RDEU	16	20	40	24	2	0

*Number shows in above tables indicates how many participants' data can be explained by the corresponding model best.

The EU theory seems to be the best in explaining data whether in terms of AIC or BIC. For AIC, the RDEU theory is the second best, then, in order, MM-RDEU, MM-EU, AMM-EU. AMM-RDEU is the worst. For BIC, *excluding EU*, models were ranked from the best to worst as follows: MM-EU, RDEU, MM-RDEU, AMM-EU and AMM-RDEU. The results are reasonable. When n is large, BIC's penalty $\ln(n)k$ supports simpler models. Compared with it, AIC balances goodness-of-fit and model complexity with a gentler penalty $2k$, so the MM-EU model was slightly better than the RDEU theory in the BIC rankings. Also, the AMM model was the worst for both ranks, so I can infer that most participants did not consider the probabilities as ambiguous. Figure 6 indicates that a majority of participants tend to be fully optimistic (i.e. estimated $\alpha = 1$), and such a distribution pattern also reflects that subjects did not treat this valuation problem in a more complex way separating it into the worst condition and the best condition. As the EU model emerges as the best, it can be inferred that subjects simply subjectively assigned the probability of the largest possible outcome despite the fact that most of them were risk averse (see Figure 7). These estimated results located within the reasonable range and consisted with estimations of previous studies which also applied the BDM method to elicit subject's attitudes towards risk (e.g. James, 2007; Zhou and Hey, 2018).

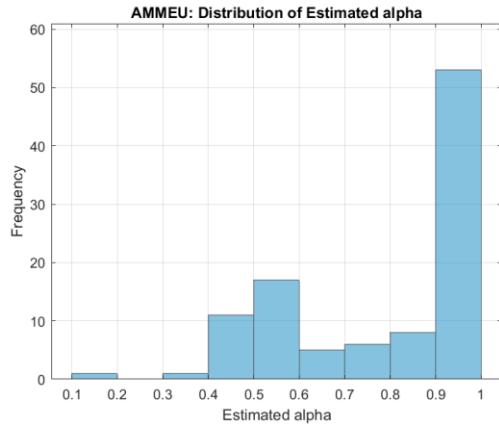


Figure 6. Distribution of estimated α under the AMM-EU model

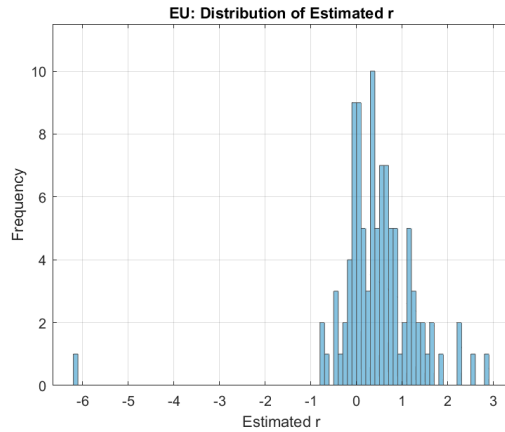


Figure 7. Distribution of estimated r under the EU model

For EU, RDEU and MM-EU models, there were 66, 47 and 75 participants respectively perceived the probability of O_4 as greater than q , the real probability. The distribution of estimated q can be seen in Figure 8. Less than half of RDEU participants made decisions with a larger perceived probability q , but according to Figure 9, a majority of estimated g , the parameter of the weighting function that indicates the degree of probability distortion, was greater than 1, so most of the participants underweighted large probabilities and overweighted large probabilities. This finding can also support why the RDEU model was the second-best in the AIC rank.

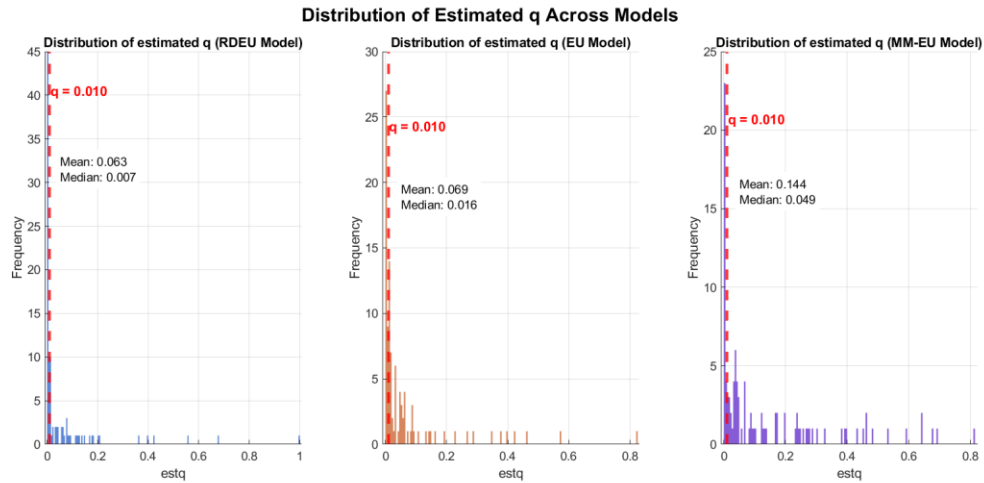


Figure 8. Distribution of estimated q for the EU, RDEU and MM-EU models.

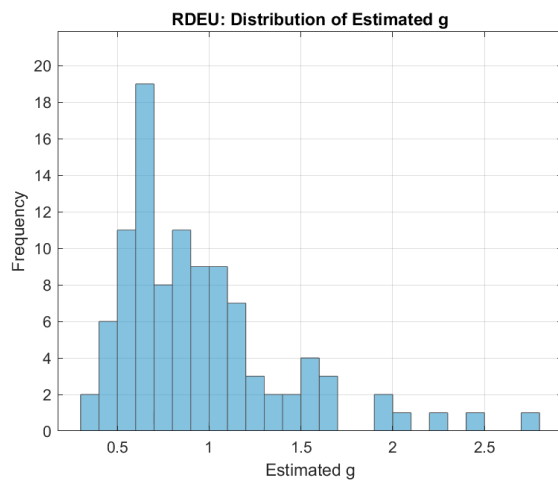


Figure 9. The distribution of estimated g

It is also understandable that the EU is the best-fitting model for this experiment with following points: First, the blind box transaction is a single-stage, gain-framed lottery with transparent probabilities. Subjects are not presented with complex multi-outcome prospects, mixed gains and losses, or ambiguous probabilities in where objective probabilities were not informed to subjects directly. And under such contexts, more complex models typically demonstrate their added value. Second, I hypothesise that consumers engage with this market in a relatively straightforward, heuristic, manner ('I might get a rare one'), rather than performing intricate mental calculations. The EU model, as the simplest model, provides the most direct tool to measure the baseline deviation from risk-neutrality in this simple environment. In other words, individuals may not use more complicated way to process this simple consumption procedure.

Since EU emerges as the best-fitting model, it should be further investigated. By the Wald Test, with EU, there were 12 participants with a significantly-larger-than- q_0 value of q . Among them, there were 5 participants disclosing risk-seeking attitudes in Part 1 with a negative r . Most of participants were

risk averse or close to risk neutral. Detail is shown in Table 2. Data was divided into two groups: *Group 1* contains these 12 participants' data and estimated results; *Group 2* contains all information on those participants whose perceived probability was *less* than $q_0 = 0.01$.

Noted that the time taken to answer to Part 2, denoted by t_2 , may well influence their decisions, I looked at the relationship between their estimate of q and t_2 with the following regressions. When the estimated q is regressed on t_2 with an intercept, the following results is obtained (t-stats in parentheses):

For Group 1:

$$q = 0.43497 - 0.00022623t_2$$

(1.988) (-0.62)

12 observations, 10 Error degrees of freedom, Root Mean Squared Error=0.237

For Group 2:

$$q = -0.00079764 + 5.7773e-06t_2$$

(-0.52174) (2.4666)

36 observations, 34 Error degrees of freedom, Root Mean Squared Error = 0.00337

There is a close relationship between the total time taken in Part 2 and the estimated q for those who had an estimated q *lower* than the real one. This relationship suggests that for subjects in Group 2 if they take a longer time for each problem in Part 2, they would get closer to the true value q_0 . However, for those in Group 1, the total time taken in Part 2 did not significantly affect their probability perception. The significant positive intercept disclosed that there were other factors that should be considered, and they led to a higher q .

A regression of the noise in the estimation (the standard error) in Part 2 on t_2 and t_2^2 was estimated too:

For Group 1:

$$se = 0.00015034t_2 - 1.1766e-07t_2^2$$

(2.5429) (-1.4947)

12 observations, 10 Error degrees of freedom, Root Mean Squared Error = 0.0411

For Group 2:

$$se = 0.0012053t_2 - 9.1232e-08t_2^2$$

(4.9936) (-0.35041)

36 observations, 34 Error degrees of freedom, Root Mean Squared Error = 0.393

For both groups, t_2 significantly, positively and linearly affected se . It appears that the longer the time took, the more thoughts subjects might have, and they might think even too much. Their stated values would have a larger deviation from their true valuations (i.e. the Certain Equivalents) with larger noise. Combined with the result of the first regression displayed, subjects coming from Group 2 might have

realized what the real probability of O_4 happening was as time went by, but it was possible that they thought it too complicated and ended up not reporting their true valuations.

Subject No.	r	q	se	Z	p-value	Loglikelihood in Part 2
7	0.388	0.266	0.030	8.393	0.000	-98.183
8	1.631	0.465	0.047	9.680	0.000	-89.545
11	0.025	0.110	0.018	5.680	0.000	-109.202
15	-0.110	0.290	0.017	16.036	0.000	-109.358
22	-0.258	0.134	0.032	3.824	0.000	-96.948
26	0.181	0.143	0.028	4.793	0.000	-100.027
33	-0.791	0.084	0.022	3.390	0.000	-105.003
56	2.252	0.820	0.040	20.394	0.000	-92.892
65	1.824	0.350	0.129	2.641	0.004	-69.402
67	-0.492	0.399	0.122	3.198	0.001	-70.544
74	-0.094	0.052	0.024	1.774	0.038	-103.033
96	1.105	0.572	0.017	32.445	0.000	-109.484

Table 2. The result of the Wald Test ($H_0: q = 0.01$; $H_1: q > 0.01$ with the significance level $\alpha = 0.05$)

8. Conclusion

This experiment was aimed at understanding blind box consumption behaviour, trying to explain it from the probability misperception perspective via a laboratory experiment. Producers of blind boxes always introduce a ‘secret edition’ or ‘hidden edition’ as the largest possible outcome of a blind box with very low probability. As they feature rareness, these kinds of figures are suggested to be more valuable and are traded at much higher prices in the second-hand market like e-bay compared with the regular figures. The experiment investigated whether individuals misjudge the probability of getting a ‘secret edition’, overvaluing the blind box, and thus purchase blind boxes and appear to exhibit risk-seeking behaviour.

Instead of real blind boxes, lotteries were used in the experiment. This setting excluded the influence of the blind boxes’ collectable feature, and avoided the problem that within a series of blind boxes, the value of different figurines varies from person to person. It would be much more complex to elicit participants’ valuation for each possible figurine first. The implementation of the lottery (with 3 of the outcomes equally valuable), enables me to avoid having to rank the outcomes when applying RDEU. Additionally, my experiment was a one-shot individual decision-making experiment, and consequently it ruled out possible effects of the Availability Heuristic and the Gambler’s Fallacy. The BDM method was the centre of the experiment. Participants were asked to value a series of three-possible-outcome lotteries in Part 1 and value a set of four-possible-outcome lotteries in Part 2 via this method. The data collected in Part 1 served for risk preference elicitation, and data collected in Part 2 was used for inferring participants’ perceived probability of the largest possible outcome of lotteries in Part 2.

Four theoretical models: EU, RDEU, the MM model and the AMM model were fitted to the data from the experiment. According to both the AIC and the BIC results, EU fitted the data best. Sixty-six (65%) of participants perceived the probability of the largest possible outcome *greater* than its objective probability, and thus overvalued lotteries; the remaining 36 (35%) perceived the probability as being *lower*, and hence undervalued lotteries. Among the 66 overestimating the probability, there were 12 participants whose subjective probabilities were *significantly higher* than the real probability. This bias persisted even under full transparency, indicating that mere disclosure of odds is insufficient to correct misperception. Let me look at the time taken to reach a decision, as this may indicate something about their commitment to the experiment. If these 12 participants (those whose subjective probabilities were *significantly higher* than the real probability) are compared to those who assigned a lower probability to the largest possible outcome, the total time taken in Part 2 only affected those who assigned a lower probability of their perception of q . Moreover, while participants who perceived a smaller probability took a longer time to make decisions, their subjective probability was closer to the true one. However, the time taken to reach decisions is only one of other factors that are waiting to be investigated in further studies.

These findings clarify the economic mechanism behind the blind box phenomenon: its appeal is not merely due to collectibility or ambiguity, but to a deep-seated tendency to overweight low-probability gains. This explains the commercial success of strategies such as Pop Mart's "secret edition" marketing and the high secondary-market premiums for rare items.

More importantly, this research extends beyond entertainment markets. It offers a direct explanation for parallel behavioral puzzles in protective markets such as insurance. The same probability-weighting bias that drives overvaluation of blind boxes likely contributes to the undervaluation of low-probability, high-impact risks like floods or disability where individuals neglect coverage despite accessible statistical information. In both contexts, objective probabilities are cognitively overridden, whether in the pursuit of a rare gain or the dismissal of a rare loss.

Overall, these results challenge the sufficiency of informational remedies in consumer protection policy. Just as mandated odds disclosure in loot box regulation may not prevent overpayment, simply providing insurance probability data may not close protection gaps. Effective policy must address the underlying cognitive bias, potentially through behavioral nudges that make low-probability risks more salient or through structural safeguards that limit extreme probability exploitation in commercial lottery products. For future studies, they should test debiasing interventions, such as probability training or affective framing, and explore whether insights from the design of engaging markets like blind boxes can be ethically applied to improve financial decision-making in domains such as insurance and pensions.

References

- Asioli, D., Mignani, A., & Alfnes, F. (2021). Quick and easy? Respondent evaluations of the Becker–DeGroot–Marschak and multiple price list valuation mechanisms. *Agribusiness*, 37(2), 215–234. <https://doi.org/10.1002/agr.21668>
- Becker, G. M., Degroot, M. H., & Marschak, J. (1964). Measuring utility by a single-response sequential method. *Behavioral Science*, 9(3), 226–232. <https://doi.org/10.1002/bs.3830090304>
- Bonniot-Cabanac, M., & Cabanac, M. (2009). Pleasure in decision-making situations: Politics and gambling. *Journal of Risk Research*, 12(5), 619–645. <https://doi.org/10.1080/13669870802579798>
- China Daily. (2019, November 25). *Eager collectors keep blind boxes squarely in sights*. Retrieved February 5, 2025, from https://www.chinadaily.com.cn/a/201911/25/WS5ddb1535a310cf3e35579950_1.html
- Chen, S. (2024). The Psychological Analysis of the Blind Box Craze in the Chinese Market: A Case Study of POP MART. *Journal of Education, Humanities and Social Sciences*, 43, 128–133. <https://doi.org/10.54097/135s5t77>
- Dave, C., Eckel, C. C., Johnson, C. A., & Rojas, C. (2010). Eliciting risk preferences: When is simple better? *Journal of Risk and Uncertainty*, 41(3), 219–243. <https://doi.org/10.1007/s11166-010-9103-z>
- Ghirardato, P., Maccheroni, F., & Marinacci, M. (2004). Differentiating ambiguity and ambiguity attitude. *Journal of Economic Theory*, 118(2), 133–173. <https://doi.org/10.1016/j.jet.2003.12.004>
- Gilboa, I., & Schmeidler, D. (1989). Maxmin expected utility with non-unique prior. *Journal of Mathematical Economics*, 18(2), 141–153. [https://doi.org/10.1016/0304-4068\(89\)90018-9](https://doi.org/10.1016/0304-4068(89)90018-9)
- Gong, X., Yee, C. L., Lee, S. Y., Saif, A. N. M., Liu, M., & Anonhi, F. (2024). Unveiling the enigma of blind box impulse buying curiosity: The moderating role of price consciousness. *Heliyon*, 10(24), e40564. <https://doi.org/10.1016/j.heliyon.2024.e40564>
- Hertwig, R., Barron, G., Weber, E. U., & Erev, I. (2004). Decisions from Experience and the Effect of Rare Events in Risky Choice. *Psychological Science*, 15(8), 534–539.
- Hey, J. D., & Orme, C. (1994). Investigating Generalizations of Expected Utility Theory Using Experimental Data. *Econometrica*, 62(6), 1291–1326. <https://doi.org/10.2307/2951750>
- Holt, C. A., & Laury, S. K. (2002). Risk Aversion and Incentive Effects. *The American Economic Review*, 92(5), 1644–1655.
- James, D. (2007). Stability of risk preference parameter estimates within the Becker-DeGroot-Marschak procedure. *Experimental Economics*, 10(2), 123–141. doi:10.1007/s10683-006-9136-y
- Kahneman, D., & Tversky, A. (1979). Prospect Theory: An Analysis of Decision under Risk. *Econometrica*, 47(2), 263–291. <https://doi.org/10.2307/1914185>
- Klibanoff, P., Marinacci, M., & Mukerji, S. (2005). A Smooth Model of Decision Making under Ambiguity. *Econometrica*, 73(6), 1849–1892. <https://doi.org/10.1111/j.1468-0262.2005.00640.x>
- Loomes, G. (1991). Evidence of a New Violation of the Independence Axiom. *Journal of Risk and Uncertainty*, 4(1), 91–108.

- Maccheroni, F., Marinacci, M., & Rustichini, A. (2006). Ambiguity Aversion, Robustness, and the Variational Representation of Preferences. *Econometrica*, 74(6), 1447–1498.
- Mvondo, G. F. N., Jing, F., & Hussain, K. (2023). What's in the box? Investigating the benefits and risks of the blind box selling strategy. *Journal of Retailing and Consumer Services*, 71, 103189. <https://doi.org/10.1016/j.jretconser.2022.103189>
- Phan, T. A., & Hoai, T. T. (2025). Chasing the scarcity: How fear of missing out and motivations drive willingness to pay in collectible markets. *Journal of Marketing Communications*, 1–15. <https://doi.org/10.1080/13527266.2025.2461143>
- Quiggin, J. (1993). *Generalized Expected Utility Theory*. Springer Netherlands. <https://doi.org/10.1007/978-94-011-2182-8>
- Schmeidler, D. (1989). Subjective Probability and Expected Utility without Additivity. *Econometrica*, 57(3), 571–587. <https://doi.org/10.2307/1911053>
- Tversky, A., & Kahneman, D. (1971). Belief in the law of small numbers. *Psychological Bulletin*, 76(2), 105–110. <https://doi.org/10.1037/h0031322>
- Tversky, A., & Kahneman, D. (1973). Availability: A heuristic for judging frequency and probability. *Cognitive Psychology*, 5(2), 207–232. [https://doi.org/10.1016/0010-0285\(73\)90033-9](https://doi.org/10.1016/0010-0285(73)90033-9)
- Tversky, A., & Kahneman, D. (1992). Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5(4), 297–323. <https://doi.org/10.1007/BF00122574>
- Xia, F., Xu, Y., Zhang, H., & Yuan, X. (2025). The effect of doll blind box uncertainty on consumers' irrational consumption behavior: The role of instant gratification, Gambler's fallacy, and perceived scarcity. *BMC Psychology*, 13, 332. <https://doi.org/10.1186/s40359-025-02644-w>
- Yahoo Finance. (2025). *Pop Mart International Group Limited (9992.HK) income statement* [Dataset]. Retrieved March 4, 2025, from <https://uk.finance.yahoo.com/quote/9992.HK/financials/>
- Zhou, W., & Hey, J. (2018). Context matters. *Experimental Economics*, 21(4), 723–756. doi:10.1007/s10683-017-9546-z
- Zhang, Y., Zhou, H., & Qin, J. (2022). Research on the effect of uncertain rewards on impulsive purchase intention of blind box products. *Frontiers in Behavioral Neuroscience*, 16, 946337. <https://doi.org/10.3389/fnbeh.2022.946337>

Technical Appendix

1. Expected Utility (EU) Theory

The calculation of CE by the standard EU model for both 2 parts is shown below:

Part 1:

$$EU = u(O_1)p_1 + u(O_2)p_2 + u(O_3)p_3$$

$$u(CE) = EU$$

$$\frac{CE^{1-r}}{1-r} = EU$$

$$CE = (EU(1-r))^{1/(1-r)}$$

Part 2:

$$EU = p[u(O_1) + u(O_2) + u(O_3)] + qu(O_4)$$

$$u(CE) = EU$$

$$\frac{CE^{1-r}}{1-r} = EU$$

$$CE = (EU(1-r))^{1/(1-r)}$$

2. Rank Dependent Expected (RDEU) Theory

The RDEU theory originally was proposed by Quiggin in 1982, and it was further developed and introduced by Tversky and Kahneman (1992) as a derivative of the Prospect Theory. Under certain restrictions, it reduces to Cumulative Prospect Theory. RDEU assumes that individuals rank possible outcomes in increasing (or decreasing) order and weigh them with distorted probabilities. To calculate the Rank Dependent Expected Utility of a lottery, possible outcomes are ranked from the worst to

best. The weighting function applied, $w(p) = \frac{p^g}{(p^g + (1-p)^g)^{1/g}}$, was proposed by Tversky and

Kahneman (1992) with the parameter g that indicates the degree of probability distortion. When $g=1$, there is no distortion, and the model reduces to EU. If $g < 1$, the probability function displays an inverse S-shaped curve, which means that the individual overweighted small probabilities and underweighted large probabilities.

In Part 1, each lottery has three possible outcomes: O_1 , O_2 and O_3 , and $O_1 < O_2 < O_3$ with corresponding probabilities: p_1 , p_2 and p_3 , and the CE can be calculated as follows:

$$RDEU = u(O_1)w(p_1) + u(O_2)[w(p_1 + p_2) - w(p_1)] + u(O_3)[1 - w(p_1 + p_2)]$$

$$u(CE) = RDEU$$

$$\frac{CE^{1-r}}{1-r} = RDEU$$

$$CE = (RDEU(1-r))^{1/(1-r)}$$

In Part 2, there are four possible outcomes: O_1, O_2, O_3 and O_4 , with $O_1 < O_2 < O_3 < O_4$. The perceived probabilities are: p_1, p_2, p_3 and q . As $p_1 = p_2 = p_3$ based on the setting of the experiment, they can be simply denoted by p , and $p = \frac{1-q}{3}$. The CE can be calculated as follows:

$$RDEU = u(O_1)w(p) + u(O_2)[w(2p) - w(p)] + u(O_3)[w(3p) - w(2p)] + u(O_4)[1 - w(3p)]$$

$$u(CE) = RDEU$$

$$\frac{CE^{1-r}}{1-r} = RDEU$$

$$CE = (RDEU(1-r))^{1/(1-r)}$$

3. MaxMin (MM) Model

The MM model that was proposed by Gilboa and Schmeidler (1989) assumes that individuals are fully pessimistic, only looking at the worst outcome and maximizing the worst Expected Utility. To estimate the parameters of this model, participants were assumed to perceive the lowest probability of O_4 happening, which is denoted as $\underline{q}$. I applied the MM model with both EU and RDEU.

3.1 MM-EU model

As $\underline{q}$ was only available in Part 2, the calculation of CE in Part 1 is as same as that by using the EU model.

For Part 2:

$$\bar{p} = \frac{1-q}{3}$$

$$MMEU = \bar{p}[u(O_1) + u(O_2) + u(O_3)] + \underline{q}u(O_4)$$

$$u(CE) = MMEU$$

$$\frac{CE^{1-r}}{1-r} = MMEU$$

$$CE = (MMEU(1-r))^{1/(1-r)}$$

3.2 MM-RDEU model

The calculation of CE in Part 1 is as same as that by using the RDEU model.

For Part 2:

$$\bar{p} = \frac{1 - \bar{q}}{3}$$

$$MMRDEU = u(O_1)w(\bar{p}) + u(O_2)[w(2\bar{p}) - w(\bar{p})] + u(O_3)[w(3\bar{p}) - w(2\bar{p})] + u(O_4)[1 - w(3\bar{p})]$$

$$u(CE) = MMRDEU$$

$$\frac{CE^{1-r}}{1-r} = MMRDEU$$

$$CE = (MMRDEU(1-r))^{1/(1-r)}$$

4. α MaxMin (AMM) model

This postulates that the DM maximises a weighted average of the worst expected utility and the best expected utility with respective weights α and $1-\alpha$. The parameter α is termed the DM's *ambiguity aversion*. When $\alpha = 1$, α MaxMin reduces to MM. In my analysis of the data from the experiment, experiment, I assumed that each subject had a *range* of perceived probabilities of O_4 with a lower bound $\underline{q}$ and an upper bound $\bar{q}$. I applied the AMM model with both EU and RDEU, and estimated $\underline{q}$ and $\bar{q}$.

4.1 AMM-EU model

As $\underline{q}$ and $\bar{q}$ were only available in Part 2, the calculation of CE in Part 1 is as same as that by using the EU model.

For Part 2:

$$\bar{p} = \frac{1 - \bar{q}}{3}$$

$$\underline{p} = \frac{1 - \underline{q}}{3}$$

$$AMMEU_{lower} = \bar{p}[u(O_1) + u(O_2) + u(O_3)] + \underline{q}u(O_4)$$

$$AMMEU_{upper} = \underline{p}[u(O_1) + u(O_2) + u(O_3)] + \bar{q}u(O_4)$$

$$u(CE) = \alpha AMMEU_{lower} + (1 - \alpha) AMMEU_{upper} = \frac{CE^{1-r}}{1-r}$$

$$CE = \alpha (AMMEU_{lower}(1-r))^{1/(1-r)} + (1 - \alpha) (AMMEU_{upper}(1-r))^{1/(1-r)}$$

4.2 AMM-RDEU model

The calculation of CE in Part 1 is as same as that by using the RDEU model.

For Part 2:

$$\bar{p} = \frac{1-q}{3}$$

$$\underline{p} = \frac{1-\bar{q}}{3}$$

$$AMMRDEU_{lower} = u(O_1)w(\bar{p}) + u(O_2)[w(2\bar{p}) - w(\bar{p})] + u(O_3)[w(3\bar{p}) - w(2\bar{p})] + u(O_4)[1 - w(3\bar{p})]$$

$$AMMRDEU_{upper} = u(O_1)w(\underline{p}) + u(O_2)[w(2\underline{p}) - w(\underline{p})] + u(O_3)[w(3\underline{p}) - w(2\underline{p})] + u(O_4)[1 - w(3\underline{p})]$$

$$u(CE) = aAMMRDEU_{lower} + (1-a)AMMRDEU_{upper} = \frac{CE^{1-r}}{1-r}$$

$$CE = a(AMMRDEU_{lower}(1-r))^{1/(1-r)} + (1-a)(AMMRDEU_{upper}(1-r))^{1/(1-r)}$$